

ADR 004 - Renaming of the DataSetMessage key 'Source' and 'Filter' to achieve OPC compliance

Status	PROPOSED
Date	Jan 14, 2026
Deciders	@David Weiss , @Blumenstock, Werner (DI FA DSP SEN TRS) , @Thomas Weinschenk , @Konrad Heidrich
Consulted	
Informed	

Context and Problem Statement

The **Source** key (see [8.1.6](#)) currently used in the DataSetMessage (s. [9.2.3](#)) is not compatible with the OPC UA specification Part 14. At the time the OI4-OEC-GL was developed, there was no equivalent for the meaning of the **Source** key in the OPC UA specification.

The **Filter** key (see [8.1.7](#)) currently used in the DataSetMessage (s. [9.2.3](#)) is not compatible with the OPC UA specification Part 14. At the time the OI4-OEC-GL was developed, there was no equivalent for the meaning of the **Filter** key in the OPC UA specification.

With the introduction of the OPC UA specification [Part 14 V1.05.04](#), additional keys were defined which could replace the **Source** key and **Filter** key with something OPC compliant.

Decision Drivers

- The OEC-GL describes the practical application of OPC UA Part 14 without implementing it in a compliant manner.
- The OI4 stands for the use of existing standards instead of reinventing the wheel.
- A standard-compliant description of the message bus payload will hopefully enable our users to reuse existing software components in the future.

Considered Options

Source corresponds to the source of the value. In the OEC context, the source is specified as an OI4Identifier.

In OPC context the `DataSetWriterName`, which is part of the [DataSetMessage](#) (UA Part 14: PubSub - 7.2.5.4) represents the source of the `Payload` best.

`DataSetWriterName` is of type string, therefore the OEC defined key `Source` can be renamed to `DataSetWriterName`. The value of type string was and will be an `OI4Identifier`.

NOTE (new key) `DataSetWriterName` = (old key) `Source` = (value) `<OI4Identifier>`

The `Filter` is of type string and contains information that depends on the `Resource` (see [8.1.5](#); e.g. `Resource = Data` $\Rightarrow$ `Filter = <Tag>`; `Resource = Config` $\Rightarrow$ `Filter = <Context.Name>`).

In OPC context the `WriterGroupName`, which is part of the [DataSetMessage](#) (UA Part 14: PubSub - 7.2.5.4) represents the `Filter` best. `WriterGroupName` and the OEC defined `Filter` are of type string, therefore `Filter` can be renamed to `WriterGroupName`. The value of it was and will be related to the used `Resource`.

NOTE (new key) `WriterGroupName` = (old key) `Filter` = (value) `<Tag>` or `<Context.Name>` or `<...>`

Both, the `DataSetWriterName` and the `WriterGroupName` are mandatory in the OEC context. Therefore, the corresponding flags must be set in [DataSetMessageContentMask](#) (UA Part 14: PubSub - 6.3.2.3.1).

In addition, the `WriterGroupName` is omitted in `NetworkMessage` if present in `DataSetMessage`, therefore it is necessary to reset the corresponding flag in [NetworkMessageContentMask](#) (UA Part 14: PubSub - 6.3.2.1.1).

For better transparency the OEC guideline shall contain default values of all `ContentMask` registers. This will make it much easier to implement a consistent “OEC interpreter”.

NOTE A good way to evaluate the different formats, depending on the `ContentMask` flags is to “play” around with the tool [OPC UA PubSub Formatter](#).

Decision Outcome/ Chosen Option

- The Key `Source` will be replaced globally with the Key `DataSetWriterName` .
- The Key `Filter` will be replaced globally with the Key `WriterGroupName` .

Consequences/ Pros and Cons

- ➖ Not backward compatible to OEC-GL v1.1.x. Not only but also because Part 14 V1.05.x is not backward compatible to v1.04.x
- ➖ Existing OEC solutions are not able to get a new compliance batch, based on v2.x of the OEC-GL, without changes.
- ➕ Compatible to newest Part 14 specification, which might make it easier to use open source components (the OI4 stands for the use of existing standards instead of reinventing the wheel).
- ➕ Consuming data from or providing data for existing OPC UA implementations should be easier

More Information

See [UA Part 14: PubSub](#) V1.05.04 from 29th of November 2024 for full information. Especially the following chapters:

- [UA Part 14: PubSub - 7.2.5.4 DataSetMessage](#)
- [UA Part 14: PubSub - 6.3.2.3.1 DataSetMessageContentMask](#)
- [UA Part 14: PubSub - 7.2.5.3 NetworkMessage containing DataSetMessages](#)
- [UA Part 14: PubSub - 6.3.2.1.1 NetworkMessageContentMask](#)

For a better understanding the tool [OPC UA PubSub Formatter](#) is recommended.